Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/18/2004
PI Name:	Wayner Jr., Peter Ph.D.		
Project Title:	R & D work on the Constrained Vapor Bubble System for a Microgravity Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	wayner@rpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6199
Organization Name:	Rensselaer Polytechnic Institute		
PI Address 1:	Isermann Department of Chemical Engineering		
PI Address 2:	110 8th Street		
PI Web Page:			
City:	Troy	**State:**	NY
Zip Code:	12180	**Congressional District:**	21
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/01/2001	**End Date:**	04/21/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Balasubramaniam, R.	**Contact Phone:**	216-433-2878
Contact Email:	RBalasubramaniam@grc.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Plawsky, Joel L (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NAG3-2383		
Performance Goal No.:			
Performance Goal Text:			

A near-term flight experiment concept was developed to pursue both equilibrium and non-equilibrium testing of the CVBHX operation. Flight-qualified apparatus and optical instrumentations are being designed and developed for a flight experiment on the International Space Station in 2005. Experimental results obtained under microgravity conditions on the International Space Station, ISS, with equipment developed by Northrop-Grumman will be compared with those obtained in the earth's environment at Rensselaer.
The microscopic intermolecular force (pressure) field, which is a function of the liquid film thickness profile, is measured using microcomputer enhanced video microscopy based on interferometry. The augmented Young-Laplace equation over a large range of film thickness is being evaluated. The overall heat conductance of the CVBHX as a function of liquid volume and heat flow rate is also being determined. Models of the transport processes in the contact line region of a CVBHX, which include the effects of liquid-solid and liquid-vapor intermolecular forces, have already

Task Description:	been developed. As part of this effort, these models are being further refined and the transport characteristics of the CVBHX are obtained by comparing the experimental data to numerical solutions of the model. The vertical CVBHX was found to be an ideal experimental setup in which to study the drop-wise condensation process and spreading using ethanol and propanol. Therefore, basic research on the region of drop-wise condensation is also being emphasized. The objective of this effort is to better understand the fundamental physics of evaporation and condensation as they affect the heat transfer processes in a constrained vapor bubble heat exchanger (CVBHX). This CVBHX consists of a small enclosed container partially filled with a liquid. When a temperature difference is applied to the ends of the CVBHX, evaporation occurs at the hot end and condensation at the cold end resulting in a very effective heat transfer device with great potential in space applications. The use of interfacial phenomena leads to a more natural energy transfer process. The dimensions of the system are optimized for microgravity conditions. This study is synergistic in that the space program needs passive systems like the CVBHX for thermal control and research on a low capillary pressure CVBHX requires a microgravity environment. This study is also multifaceted: (1) It is a basic study in interfacial physics because the CVBHX is used to measure the interfacial intermolecular force field; (2) It is a basic study in fluid physics because it evaluates fluid flow in ultra-thin liquid films and menisci; (3) It is a dynamic study in thermal transport because it evaluates the heat transfer capabilities of a small scale passive heat exchanger; and (4) The CVBHX can also be called a wickless heat pipe heat exchanger. The results from this study are not only applicable to space applications, for which it is optimized, but can also help the design of conventional heat pipes, which are poorly understood. NOTE THIS IS PROJECT CONTINUES WITH GRANT NNC05GA27G; see that task for subsequent reports.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although this project is primarily directed at developing heat transfer equipment for use under microgravity conditions, all the basic information obtained on passive transport processes controlled by interfacial phenomena will be of general use to enhancing our understanding of interfacial phenomena. An example of one application is the enhancement of our understanding of heat pipes in general. In addition, the optical techniques, IAI, developed under this grant will be of general use in basic research.
Task Progress:	We are working with a new Project Scientist, R. Balasubramanian, who is assisted by Sang Young Son, and a NASA Projects Team at the Glenn Research Center headed by Sue Motil on the design of an experimental system for the International Space Station. The development of the equipment for the flight experiment is coordinated with Northrop-Grumman. John Eustace is the CVB Lead at Northrop-Grumman for the design and development of the CVBHX cell which will fit into a slot on the Light Microscope Module, LMM, Good progress is being made. The Critical Design Review, CDR, for the Light Microscope Module being developed by Northrop-Grumman was held on December 10-11, 2003. We are scheduled to run the experiment on the international Space Station during the year 10/05 - 9/06. Experimental results obtained under microgravity conditions will be compared with those obtained at Rensselaer. Basic research at Rensselaer continues on the experimental and theoretical details associated with phase change processes in the CVBHX. The extensive results of our current research are presented in the nineteen (19) papers and publications listed. Eighteen (18) presentations have been given. Briefly, evaporation/condensation data from both vertical and horizontal CVB systems were obtained and analyzed for both polar (wetting) and apolar (partially wetting) fluids. The vertical system is axi-symmetric, but strongly effected by gravity. Whereas, the horizontal system is asymmetric but weakly effected by gravity. Therefore, there will be significant differences in the operation of the cell in the earth's environment versus the operation under microgravity conditions. The CVBHX was found to be an ideal experimental setup in which to study the effects of interfacial phenomena on both the evaporation and drop-wise condensation processes. Therefore, basic research on the region of drop-wise condensation is also being emphasized along with the evaporation process. The optical technique (Image Analyzing Interferometry, IAI), which is based on the measurement and analysis of the reflectivity pattern of a thin film, was significantly improved over the past year. The accuracy of the IAI system is of critical importance to the success of the mission because it is used to measure the details of the pressure field in the liquid. The accuracy was found to be excellent and various publications/presentations documenting these new results were/are-being written. Significant new results were also obtained for the oscillating contact line region. Results documenting these studies were either published or accepted for publication.
Bibliography Type:	Description: (Last Updated: 09/25/2012)
Abstracts for Journals and Proceedings	Gokhale SJ, Plawsky JL, Wayner PC Jr, DasGupta S. "Experimental Evaluation of Interfacial Profile and Pressure Variation in a Spreading Dropd During Condensation." 2003 AIChE Annual Meeting, San Francisco, CA, November 16-21, 2003. 2003 AIChE Annual Meeting, San Francisco, CA, November 16-21, 2003. Extended Abstract # 244d. , Nov-2003
Articles in Peer-reviewed Journals	Gokhale SJ, Plawsky JL, Wayner PC, DasGupta S. "Inferred pressure gradient and fluid flow in a condensing sessile droplet based on the measured thickness profile." Physics of Fluids. 2004 Jun;16(6):1942-55. http://dx.doi.org/10.1063/1.1718991 , Jun-2004
Articles in Peer-reviewed Journals	Zheng L, Plawsky JL, Wayner PC Jr, DasGupta S. "Stability and oscillations in an evaporating corner meniscus." ASME Journal of Heat Transfer. 2004 Apr;126(2):169-78. http://dx.doi.org/10.1115/1.1652046 , Apr-2004
Papers from Meeting Proceedings	Panchamgam SS, Gokhale SJ, Plawsky JL, DasGupta,S, Wayner PC Jr. "A Study of an Oscillating Corner Meniscus with Phase Change Using Image Analyzing Interferometry." 2004 ASME Heat Transfer/Fluids Engineering Summer Conference, Charlotte, NC, July 11-15, 2004. Proceedings of the 2004 ASME Heat Transfer/Fluids Engineering Summer Conference, Charlotte, NC, July 11-15, 2004.Paper no. HT-FED2004-56146, p. 317-326. http://dx.doi.org/10.1115/HT-FED2004-56146 , Jul-2004

Presentation	Gokhale, S. J.; DasGupta,S.; Plawsky, J. L.; Wayner, P. C. Jr. "Experimental Evaluation of Interfacial Profile and Pressure Variation in a Spreading Dropd During Condensation" 2003 AIChE Annual Meeting Extended Abstract # 244d, San Francisco, CA. Nov-2003
Presentation	Plawsky, J.L.; Wayner, P. C. Jr. "Constrained Vapor Bubble" Conference-Workshop on Strategic Research to Enable NASA's Exploration Mission, June 22-23, 2004, Cleveland, OH Jun-2004
Presentation	Panchamgam,S. S.; Gokhale, S. J.; DasGupta,S.; Plawsky, J. L. ;Wayner, P. C. Jr. "A Study of an Oscillating Corner Meniscus with Phase Change Using Image Analyzing Interferometry" 2004 ASME Heat Transfer/Fluids Engineering Summer Conference , Charlotte, NC, July 11-15, 2004 Jul-2004